

Homework 9

Stability Margins

1. Define the following: gain margin, phase margin, gain crossover frequency, phase crossover frequency. Make a bode plot and show these visually.
2. What does a negative gain margin look like on a bode plot? What does it imply about stability?
3. What does a negative phase margin look like on a bode plot? What does it imply about stability?
4. 6.27. Use Matlab for bode, root locus, Nyquist, and routh array methods.

Compensation

5. Briefly explain lead, lag, and lead-lag compensation. When would they be appropriate to use? How do they correspond to PI, PD, and PID controllers?
6. 6.48a only. Make $\omega_{gc} = 1$. Use Matlab for bode plots. What is the expected damping ratio for a PM of 30° ? Plot the step response for the uncompensated system $G/(1 + G)$ and the compensated system $C_{lead}G/(1 + C_{lead}G)$.

Optional Bonus

Consider the antenna subsystem plant in Problem 3, Homework 7.

7. Make a bode plot of your PI controller (part j).
8. Use this plot to help you design a compensator that meets all the requirements. Explain your methodology. Note that the lag compensator does not change the system type, so you will not be able to meet the steady state error requirement (perhaps try to keep the steady state error low, or add a $1/s$ block after the compensator and design as if it is part of the plant).
9. Show the bode plots of the PI controller and compensator on the same axes.
10. Show the simulation output from using the PI controller and lag compensator on the same axes. Are there notable differences?